
SIMBAD POST-PROCESSING PROGRAM

FOREWORD

THE SIMBAD POST-PROCESSING PROGRAM PROVIDES, FROM THE SIMBAD CALCULATION RESULTS:

- **PLOTS OF TIME SERIES OF MOORING LINE AND FENDER LOADS**
- **PLOTS OF MOTIONS, SPEEDS AND ACCELERATIONS AT THE CENTER OF GRAVITY AND AT OTHER POSSIBLE POINTS OF THE SHIP**

THESE RESULTS CAN BE ALSO POST-PROCESSED, IN ORDER TO OBTAIN:

- **TABLES OF RESULTS, INCLUDING:**
 - **MINIMUM AND MAXIMUM VALUES OVER THE TIME SERIES, F1/3 AND F1/10 VALUES THAT ARE THE MEAN OF 3RD OR 10TH HIGHEST PART OF THE TIME SERIES DISTRIBUTION**
 - **THE MOST PROBABLE MAXIMUM AND MINIMUM (MPM) STATISTICAL VALUES OF THE PEAK RESPONSES GIVEN RESPECTIVELY BY: AVERAGE VALUE + OR – 3.8 X STANDARD DEVIATION**
 - **A 3D ANIMATION BASED ON THE VPYTHON ENVIRONMENT, AS WELL AS A VIDEO CREATED FROM IMAGES OBTAINED WITH THE ANIMATION**
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THE INPUT DATA ARE READ FROM THE SIMBAD OUTPUT FILES: “*.RES”

RESULTS ARE WRITTEN INTO DIFFERENT OUTPUT FILES:

- **“MAX_MIN.TXT”, FOR MOORING LINE AND FENDER LOADS**
- **“MOTIONCG.TXT”, FOR THE MOTIONS, VELOCITIES OR ACCELERATIONS OF CENTRE OF GRAVITY**
- **“MOTIONPT.TXT”, FOR THE MOTIONS, VELOCITIES OR ACCELERATIONS OF A SPECIFIC POINT**

THESE FILES ARE DESCRIBED IN DETAILS HEREAFTER.

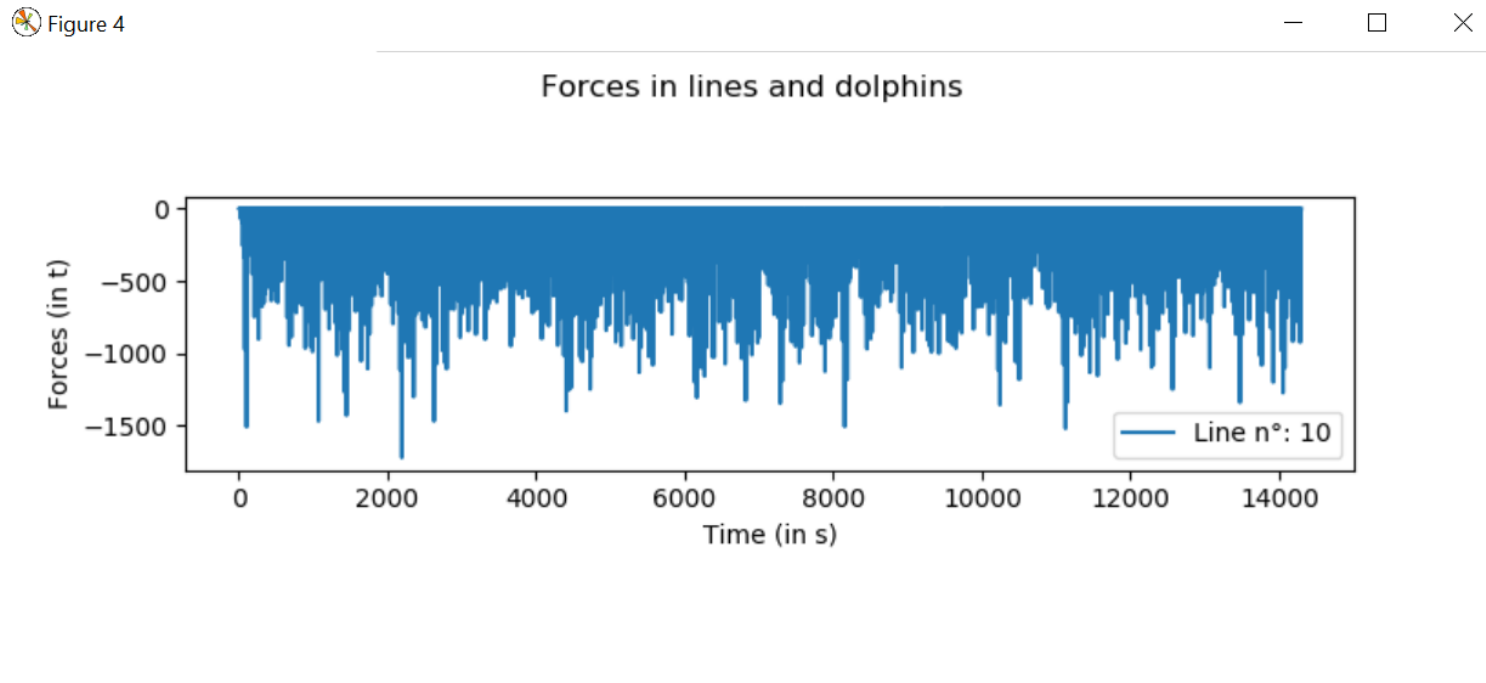
RUNNING THE SIMBAD POST-PROCESSING PROGRAM: AN EXAMPLE OF RUN IS GIVEN HEREAFTER

```
1: Tensions in lines,  
2: Motions of CG,  
3: Velocities of CG,  
4: Accelerations of CG,  
5: Motions of a particular point,  
6: Velocities of a particular point,  
7: Accelerations of a particular point  
1
```

```
Reading the file results in the LINES.RES file  
Time start (in s), default = 400s:  
Time end (in s), default = 14300s:
```

CHOICE OF PARAMETER TO BE PLOTTED AND ANALYSED

DETERMINATION OF TIME START AND END FOR THE PLOTS AND ANALYSIS. GENERALLY, THE TIME START IS THE RAMP-UP TIME INTRODUCED IN THE SIMBAD PROGRAM AND THE TIME END THE LENGTH OF THE TIME SERIES. NEVERTHELESS, IT IS RECOMMENDED TO CUT OFF THE LAST 100 s AS THIS LAST TIME PERIOD MAY BE AFFECTED BY NUMERICAL INCONSISTENCIES.



RUNNING THE SIMBAD POST-PROCESSING PROGRAM: RESULT FILE “MAX_MIN.TXT”, FOR MOORING LINE AND FENDER LOADS

Line n°:	Forces					
	Min	Max	Minst.	Maxst.	F1/3	F1/10
1	0.00	112.10	0.00	100.65	65.15	77.77
2	0.00	118.40	0.00	107.35	68.84	82.36
3	0.00	113.80	0.00	102.40	66.07	78.82
4	17.99	42.29	19.85	38.59	32.02	34.48
5	16.55	50.38	19.43	46.11	36.81	40.20
6	0.00	153.30	0.00	133.53	84.34	101.36
7	0.00	158.60	0.00	138.25	86.85	104.66
8	0.00	151.50	0.00	132.54	83.51	100.69
9	-1521.00	0.00	-1485.82	0.00	-813.07	-1013.06
10	-1720.00	0.00	-1642.66	0.00	-890.24	-1125.18

*MINIMUM AND MAXIMUM VALUES, MAXIMUM AND
MINIMUM STATISTICAL VALUES, F1/3 AND F1/10 VALUES*

**RUNNING THE SIMBAD POST-PROCESSING PROGRAM:
RESULT FILE “MOTIONCG.TXT”, FOR THE MOTIONS, VELOCITIES OR ACCELERATIONS OF
CENTRE OF GRAVITY**

Type:	Motion		Minst.	Maxst.	F1/3	F1/10
	Min	Max				
SURGE	-0.049	0.0724	-0.059	0.0857	0.0458	0.0567
SWAY	-2.676	1.12	-2.494	1.1784	-1.274	-1.668
HEAVE	-1.281	1.381	-1.264	1.2610	-0.671	-0.857
ROLL	-3.81	3.443	-3.754	3.4590	-2.010	-2.520
PITCH	-0.034	0.0414	-0.035	0.0410	0.0220	0.0274
YAW	-0.494	0.3937	-0.514	0.3927	-0.268	-0.335

*MINIMUM AND MAXIMUM VALUES, MAXIMUM AND MINIMUM
STATISTICAL VALUES, F1/3 AND F1/10 VALUES*

RUNNING THE SIMBAD POST-PROCESSING PROGRAM:
RESULT FILE “MOTIONPT.TXT”, FOR THE MOTIONS, VELOCITIES OR ACCELERATIONS OF A PARTICULAR POINT

Type:	Motion		Minst.	Maxst.	F1/3	F1/10
	Min	Max				
X	-2.772	1.215	-2.763	1.7640	-1.423	-1.761
Y	-1.156	0.5952	-0.981	0.6425	-0.556	-0.728
Z	-0.986	0.7788	-0.849	0.7255	-0.469	-0.584

MINIMUM AND MAXIMUM VALUES, MAXIMUM AND MINIMUM STATISTICAL VALUES, F1/3 AND F1/10 VALUES

RUNNING THE SIMBAD POST-PROCESSING PROGRAM:

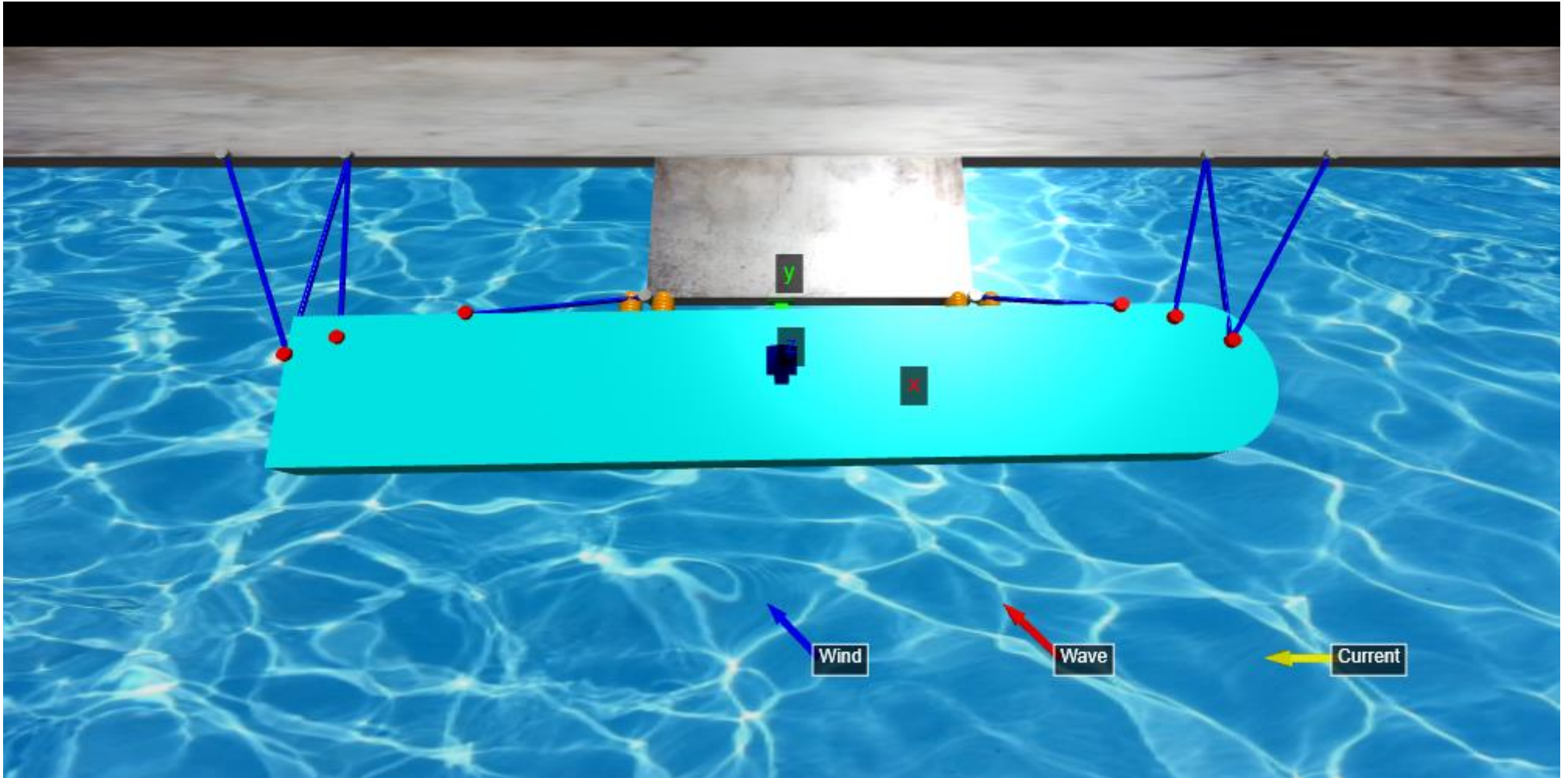
FOR THE 3D ANIMATION, THE FOLLOWING FILES ARE NECESSARY:

- THE FILE `FREQDIR.TXT` OBTAINED WITH THE LAST VERSION (V0.3) OF THE PRE-PROCESSING PROGRAM
- THE FILE « `AMAR.DAT` » OF THE MOORING SYSTEM
- THE FILE `MOVCG.RES` OF THE COG SHIP MOTION
- SOME PICTURES (IN *.JPEG) FOR THE TEXTURES OF THE SURROUNDINGS (WATER, SOIL, QUAY...)

THE ANIMATION CAN BE CHOSEN ACCORDING TO THE MOST INTERESTING TIME RANGES. HOWEVER, IT SHOULD BE NOTED THAT IF THE CHOSEN TIME RANGE IS TOO LONG, THE REPRODUCTION OF THE ROLL ROTATION MUST BE DISABLED TO AVOID INAPPROPRIATE REPRODUCTION OF THE MOVEMENTS!.

THE PRESENT ANIMATION IS GIVEN, AS AN EXAMPLE, FOR A 125,000 m³ LNG SHIP MOORED TO A JETTY AND SUBMITTED TO STRONG BOW QUARTERING WAVES ($H_s=2.5$ m, $T_p=15$ s) AND WINDS ($V=25.2$ m/s), AS WELL AS A BOW CURRENT OF 1.5 m/s. MODIFICATIONS TO THE "SMCONFIGUR.PY" MODULE WILL PROBABLY BE NECESSARY TO ADAPT IT TO OTHER SITUATIONS, WHETHER IN TERMS OF VESSEL TYPE OR MOORING CONFIGURATION.

RUNNING THE SIMBAD POST-PROCESSING PROGRAM:



VIEW OF THE 3D ANIMATION
